

CHAPTER 1

Matter, Atoms & The Periodic Table

Everything around you is made of tiny pieces called atoms.
Let's learn how they work!

 Easy Read Edition

WHAT WE WILL LEARN

- ★ What is matter?
- ★ The three types of matter
- ★ Two kinds of properties
- ★ The four states of matter
- ★ Heating and cooling
- ★ The parts of an atom
- ★ The Periodic Table
- ★ Patterns in the table
- ★ Counting atoms
- ★ Balanced equations

1.1 What Is Matter?

Look around you. Your desk. Your phone. Even the air. All of it is **matter**.

Matter is anything that has mass and takes up space.

That means almost everything! Water is matter. Rocks are matter. *You* are matter.



Water
is matter



Rocks
are matter



Air
is matter

★ Cool Fact

Every atom in your body was made inside a star. A star exploded long ago. Those pieces became you! You are made of star dust.

1.2 Three Types of Matter

All matter can be split into three big groups.

1. Elements

An element is made of just one kind of atom. Gold is an element. So is oxygen. There are about 118 known elements.

2. Compounds

A compound is made when two or more elements stick together. Water is a compound. Its parts are hydrogen and oxygen.

3. Mixtures

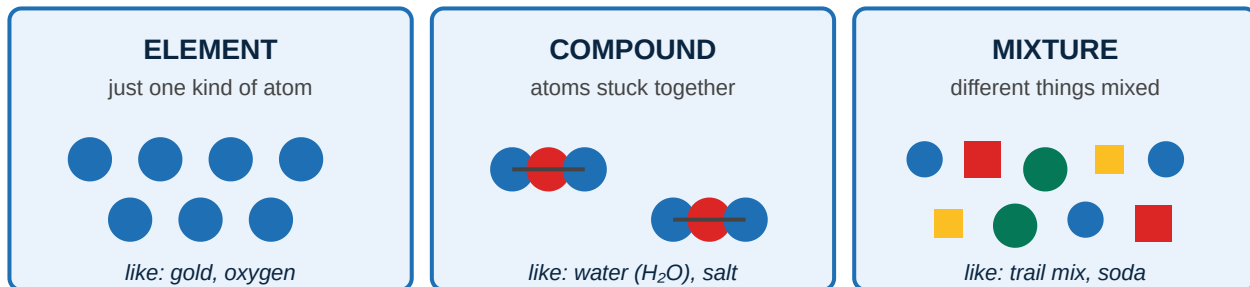
A mixture is two or more things stirred together. Trail mix is a mixture. So is salt water.

Two Kinds of Mixtures

Some mixtures look the same all the way through. These are **homogeneous** mixtures. Salt water is one. You can't see the salt.

Some mixtures show their parts. These are **heterogeneous** mixtures. A bowl of cereal is one. You can see the milk and the flakes.

Element vs. Compound vs. Mixture



Three types of matter. Atoms are the small circles!



Check Yourself

1. What is matter?
2. Is iron an element or a compound?
3. Is a salad more like a mixture or a compound?

1.3 Properties of Matter

A **property** is a fact about something. It tells you what that thing is like.

There are two types of properties.

Physical Properties

You can see a physical property. You don't have to change the thing to find it.

- **Color** — red, blue, green

- **Size** — big or small
- **Density** — how heavy for its size
- **Melting point** — when it turns to liquid
- **Boiling point** — when it turns to gas

Chemical Properties

A chemical property tells how something will *change* into a new thing.

- **Can it burn?** — Paper can burn. Rocks cannot.
- **Can it rust?** — Iron rusts. Gold does not.
- **Does it react?** — Sodium reacts with water. Sand does not.

Changes in Matter

There are also two kinds of changes.

A **physical change** changes how matter looks. But it is still the **same stuff**.

A **chemical change** makes a **new** kind of matter.

Physical Change	Chemical Change
Ice melts to water	Wood burns to ash
Paper gets cut	Iron rusts
Sugar mixes in tea	A cake bakes
Water boils to steam	Fireworks explode

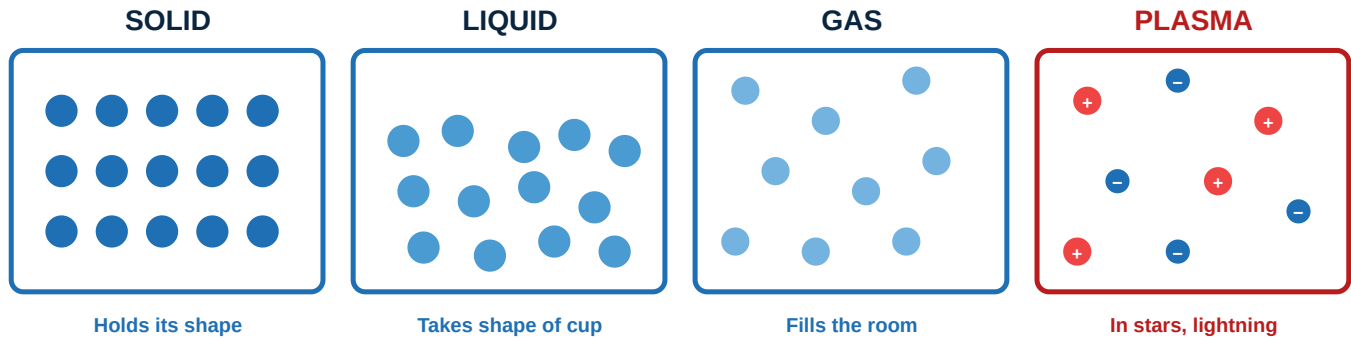


Watch Out!

Melting ice is a physical change. The ice is still water — just in a new form. Burning wood is a chemical change. The wood becomes ash and smoke — totally new!

1.4 The Four States of Matter

Matter comes in four forms. These are called **states** of matter.



The four states of matter. The dots are tiny pieces called particles.

Solid

Particles stay close. They just shake in place. Solids hold their own shape. Like ice or a brick.

Liquid

Particles are close but can slide. Liquids take the shape of their cup. Like water or milk.

Gas

Particles fly around free. Gas fills any space. Like air or steam.

Plasma

Super-hot gas with charges. Found in stars and lightning. Not on Earth much!

How Matter Changes State

Add heat → particles move faster → matter changes state.

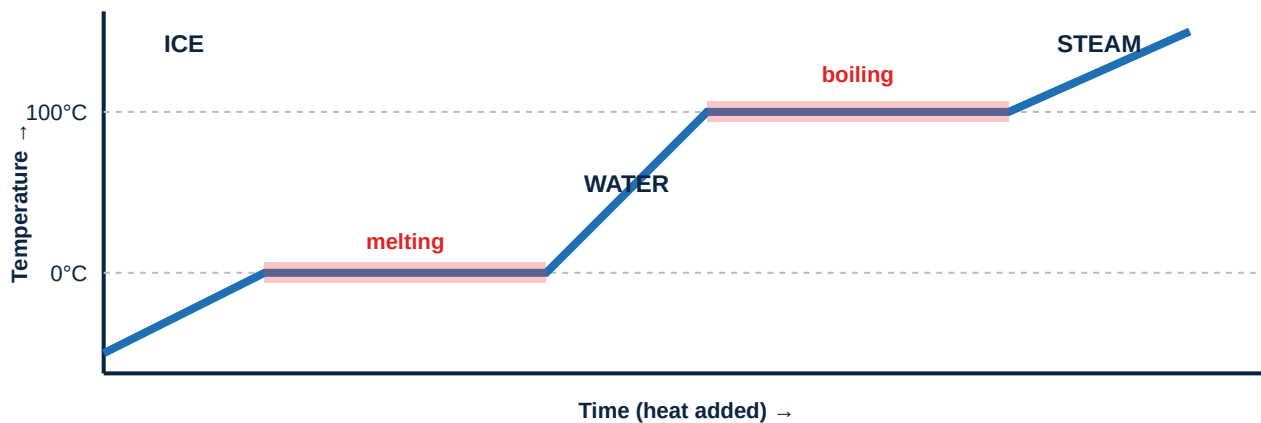
Take away heat → particles slow down → matter changes back.

1.5 Heating & Cooling Curves

Watch what happens when you heat ice.

The temperature does NOT just go up. It **stops** when the ice melts. It stops **again** when the water boils.

Why? The heat is busy breaking the ice apart. Then it goes back up.



When ice gets heat, the temperature goes up. Then it stops while ice melts. Then up again. Stops while water boils.



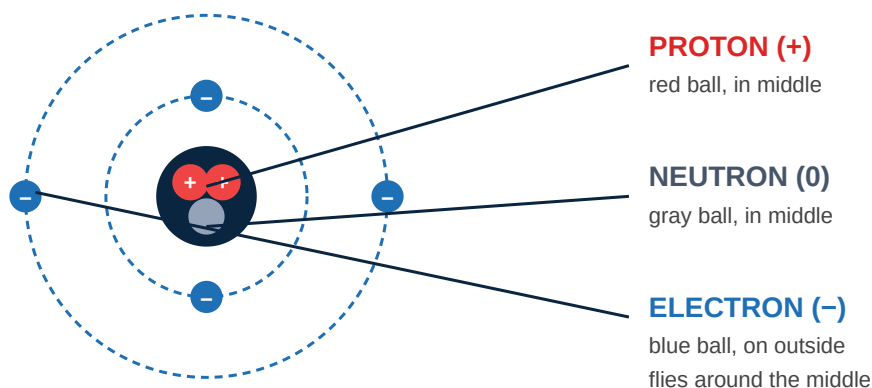
Remember

The flat parts on the graph are where the matter is changing state. The heat is being used to change the state, not warm it up.

1.6 Inside the Atom

An **atom** is the smallest piece of an element. Atoms are too small to see — even with most microscopes.

Atoms have three parts:



Inside an atom. The middle is called the nucleus. Electrons fly around it.

Part	Charge	Where?
Proton	+ (plus)	In the middle
Neutron	0 (none)	In the middle
Electron	- (minus)	Flying around

The middle part is called the **nucleus**. It holds the protons and neutrons.

Most of an atom is empty space! If the atom were as big as a football stadium, the nucleus would be the size of a pea.

1.7 The Periodic Table

There are about 118 different elements. They are all listed on a chart. This chart is called the **Periodic Table**.

Each box on the table is one element. Each box gives you facts about that element.

Atomic Number
(many protons!)

11

Na

Sodium

22.99

Symbol
short name

Element Name

Atomic Mass
(protons + neutrons)

One box from the Periodic Table. This one is for Sodium.



Easy Trick

To find the parts of an atom from the table:

- **Protons** = the atomic number
- **Electrons** = the same as protons
- **Neutrons** = atomic mass minus atomic number

For sodium: 11 protons, 11 electrons, $23 - 11 = 12$ neutrons.

1.8 Patterns in the Table

The Periodic Table is set up in **rows** and **columns**.

Periods (Rows)

A period is a row that goes across. There are 7 rows on the table.

Groups (Columns)

A group is a column that goes up and down. There are 18 columns. Elements in the same column act alike!

Three Big Groups



METALS

shiny, bend, conduct



NONMETALS

dull, break, don't conduct



METALLOIDS

a mix of both



Cool Fact

Computer chips are made of silicon. Silicon is a metalloid. That's why the tech area in California is called "Silicon Valley"!

1.9 Counting Atoms in a Formula

Chemists use short names for compounds. These short names are called **chemical formulas**.

Water is H_2O . That little 2 is a **subscript**. It tells you there are 2 hydrogen atoms.

If there is no little number, it means just 1 of that atom.

Two Rules to Count Atoms

Rule 1: Subscripts (small numbers)

A subscript only counts the letter right in front of it. In H_2O : 2 hydrogen, 1 oxygen.

Rule 2: Coefficients (big numbers in front)

A big number in front counts EVERYTHING. So $3 \text{H}_2\text{O} = 6 \text{ hydrogen} + 3 \text{ oxygen}$.

How to Count Atoms in: $2 \text{H}_2\text{SO}_4$



Count each:

$$\text{H: } 2 \times 2 = 4$$

$$\text{S: } 2 \times 1 = 2$$

$$\text{O: } 2 \times 4 = 8$$

Total: 14 atoms

How to count atoms with subscripts and coefficients.



Try It

1. How many atoms are in H_2O ?
2. How many atoms are in CO_2 ?
3. How many of each in $3 \text{H}_2\text{O}$?

1.10 Balanced Equations

A **chemical equation** shows a chemical change.

The stuff before the arrow is the **reactants**. The stuff after the arrow is the **products**.

Reactants → Products

The Big Rule



Conservation of Mass

Atoms cannot disappear. Atoms cannot appear from nothing.

The number of each type of atom must be the SAME on both sides.

When this is true, the equation is **balanced**.

A BALANCED Equation: $2 \text{H}_2 + \text{O}_2 \rightarrow 2 \text{H}_2\text{O}$

REACTANTS (left side)

H: $2 \times 2 = 4$ atoms

O: 2 atoms



PRODUCTS (right side)

H: $2 \times 2 = 4$ atoms ✓

O: $2 \times 1 = 2$ atoms ✓

Same atoms on both sides! Balanced!

A balanced equation. Same number of each atom on both sides.



Watch Out!

To balance, you can **ONLY** change the big numbers in front. **NEVER** change the little subscripts!

Changing H_2O to H_2O_2 makes a totally different thing! (H_2O is water. H_2O_2 is hydrogen peroxide!)

★ CHAPTER 1 — BIG IDEAS

- **Matter** is anything that has mass and takes up space.
- Matter can be an **element**, **compound**, or **mixture**.
- A **physical change** doesn't make new stuff. A **chemical change** does.
- Matter has four states: **solid**, **liquid**, **gas**, **plasma**.
- Atoms have three parts: **protons (+)**, **neutrons (0)**, **electrons (-)**.
- The **Periodic Table** shows all the elements in order.
- In a formula, **subscripts** count one type of atom. **Coefficients** count everything.
- **Atoms can't be made or destroyed**. That's why we balance equations.